

Chapter 13 Homework Answer Sheet

Part A Multiple choice (15 points)

1-5 ECDDDB

6-10 ABEBD

11-15 EADAD

Part B Short questions (10 points)

15. (5 points)

(a) ion-dipole

(b) dispersion forces

(c) hydrogen bonding

From weakest to strongest solvent-solute interactions: (b) < (c) < (a)

16. (6 points)

(a) $X_{\text{acetone}} = X_{\text{chloroform}} = 0.5$

For an ideal solution, Raoult's law is obeyed.

$$P_t = P_{\text{acetone}} + P_{\text{chloroform}} = 0.5 \times 360 + 0.5 \times 300 = 330 \text{ torr}$$

(b) Weak hydrogen bonds between the two molecules prevent molecules from escaping to the vapor phase and result in a lower vapor pressure, 250 torr, than an ideal solution of the same composition, 330 torr.

(c) No. Chloromethane has only one chlorine atom withdrawing electron density from the central carbon atom, the three C-H bonds are less polarized, and interactions with acetone are weaker. There is probably some vapor pressure lowering, but much less than for chloroform and acetone.

17. (4 points)

$$\Pi = iMRT,$$

here molarity = molality

molality = moles of solute / kilograms of solvent

$$\text{So, } \Pi = (i \times \text{moles of solute} \times RT) / \text{kilograms of solvent}$$

$$\text{moles of solute} = (\Pi \times \text{kilograms of solvent}) / iRT$$

$$= \frac{0.0751 \times 100.0 \times 10^{-3} \text{ kg}}{2 \times 0.0821 \text{ L} \cdot \text{atm} / \text{mol} \cdot \text{K} \times 298 \text{ K}} = 1.535 \times 10^{-4} \text{ mol}$$

The molar mass of the solute is

$$M_{\text{solute}} = \frac{m}{n} = \frac{0.036 \text{ g}}{1.535 \times 10^{-4} \text{ mol}} = 234 \text{ g/mol}$$

The molar mass of $\text{LiC}_n\text{H}_{2n+1}\text{O}_2$ is

$$6.941 + 12.01n + 1.008(2n+1) + 16.00 \times 2 = 234$$

$$\text{So } n = 14$$